

Pure Transformers are Powerful Graph Learners

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Previous work

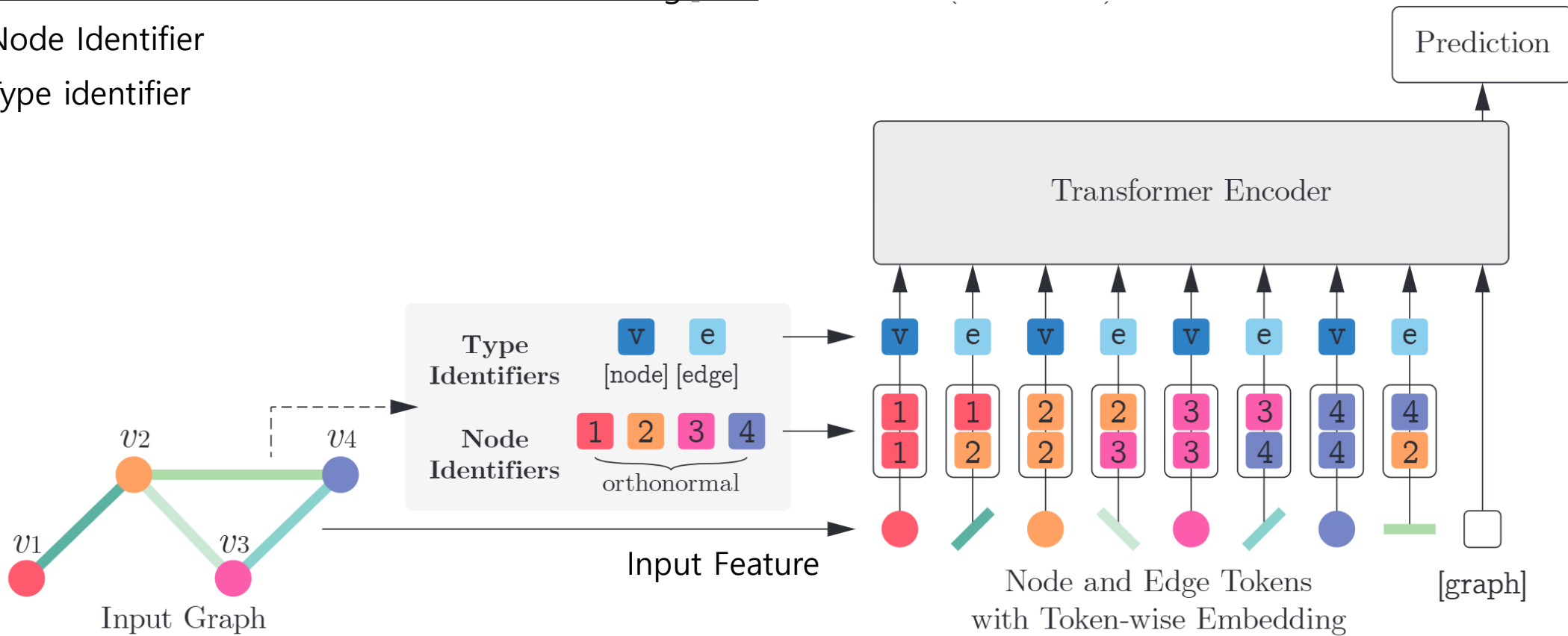
- 최근 Transformer는 자연어처리, 컴퓨터비전, 강화학습 등 다양한 머신러닝 학습 문제에서 사용됨
- 그래프 학습 분야에서도 message passing을 사용하는 GNN에 Transformer의 self-attention을 결합하려는 여러 연구를 시도함.
- 노드 전반의 Global self-attention은 그래프 구조를 반영할 수 없으므로 그래프별 아키텍처 변형을 도입함.
 - Self-attention을 local neighborhood로 제한
 - Attention bias를 통해 global self-attention에 엣지 정보를 주입

➡ **Oversmoothing 등의 한계 발생**

➡ **반대로 Transformer를 Graph에 직접 적용해보자**

TokenGT

- Transformer를 graph에 적용한 모델
- 노드 식별자와 타입 식별자로 구성된 **pure Transformer architecture**
- 그래프 구조를 표현하기 위해 Token-Wise embedding 이용
 - Node Identifier
 - Type identifier



Node Identifiers

- 입력 그래프에 주어진 연결성 구조를 표현하는 데 사용되는 정규 노드 식별자 $\mathbf{P} \in \mathbb{R}^{n \times d_p}$

For each node $v \in \mathcal{V}$, we augment the token $\mathbf{X}_v$ as $[\mathbf{X}_v, \mathbf{P}_v, \mathbf{P}_v]$.

For each edge $(u, v) \in \mathcal{E}$, we augment the token $\mathbf{X}_{(u,v)}$ as $[\mathbf{X}_{(u,v)}, \mathbf{P}_u, \mathbf{P}_v]$.

- 예를 들어, $[\mathbf{P}_u, \mathbf{P}_v][\mathbf{P}_k, \mathbf{P}_k]^\top = 1$ if and only if $k \in (u, v)$ 이고, 0이면 끝점이 아닌 것

- 정규 노드 식별자 종류

- Orthogonal random features(ORFs)** : G에 QR분해 적용해서 직교 행렬 Q, 상삼각 행렬 R의 곱으로 분해하고, Q의 각 행을 노드 식별자로 사용함.

Orthogonal random features (ORFs), e.g., rows of random orthogonal matrix $\mathbf{Q} \in \mathbb{R}^{n \times n}$ obtained with QR decomposition of random Gaussian matrix $\mathbf{G} \in \mathbb{R}^{n \times n}$ [79, 12].

- Graph Laplacian matrix** : 그래프의 인접 행렬(A)과 차수 행렬(D)로부터 정규화된 라플라시안 행렬을 계산한 뒤, 이를 고유값 분해하여 얻은 고유벡터 행렬(U)의 각 행을 노드 식별자로 사용함.

Laplacian eigenvectors obtained from eigendecomposition of graph Laplacian matrix, i.e., rows of $\mathbf{U}$ from $\Delta = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2} = \mathbf{U}^\top \mathbf{\Lambda} \mathbf{U}$, where $\mathbf{A} \in \mathbb{R}^{n \times n}$ is adjacency matrix, $\mathbf{D}$ is degree matrix, and $\mathbf{\Lambda}$, $\mathbf{U}$ correspond to eigenvalues and eigenvectors respectively [20].

Type Identifiers

- 토큰이 노드인지 엣지인지 인코딩하는 학습 가능한 타입 식별자 $\mathbf{E} = [\mathbf{E}^{\mathcal{V}}; \mathbf{E}^{\mathcal{E}}] \in \mathbb{R}^{2 \times d_e}$

For each node $v \in \mathcal{V}$, we augment the token $[\mathbf{X}_v, \mathbf{P}_v, \mathbf{P}_v]$ as $[\mathbf{X}_v, \mathbf{P}_v, \mathbf{P}_v, \mathbf{E}^{\mathcal{V}}]$.

For each edge $(u, v) \in \mathcal{E}$, we augment the token $[\mathbf{X}_{(u,v)}, \mathbf{P}_u, \mathbf{P}_v]$ as $[\mathbf{X}_{(u,v)}, \mathbf{P}_u, \mathbf{P}_v, \mathbf{E}^{\mathcal{E}}]$.

- Attention head가 특정 노드 토큰에만 집중하고 엣지 토큰은 무시하려는 경우에 중요함.

Main Transformer

- Input sequence

$$\mathbf{Z}^{(0)} = [\mathbf{X}_{[\text{graph}]}; \mathbf{X}^{in} w^{in}] \in \mathbb{R}^{(1+n+m) \times d}$$

- 노드 식별자, 타입 식별자로 증강된 토큰 feature

$$\mathbf{X}^{in} \in \mathbb{R}^{(n+m) \times (C+2d_p+d_e)}$$

- 그래프 레벨 예측을 위한 [graph] 토큰

$$\mathbf{X}_{[\text{graph}]} \in \mathbb{R}^d$$

- 학습 가능한 가중치 행렬

$$w^{in} \in \mathbb{R}^{(C+2d_p+d_e) \times d}$$

- Encoder : MSA Layer, FeedForward MLP Layer를 번갈아 쌓은 구조

Invariant Graph Network

- 그래프/하이퍼그래프의 복잡한 구조 데이터를 처리하기 위해 설계된 신경망
- Feature tensor를 입력으로 받아 스칼라 예측값을 출력함

Definition 1. An order- k Invariant Graph Network (k -IGN) is a function $F_k : \mathbb{R}^{n^k \times d_0} \rightarrow \mathbb{R}$ written as the following:

$$F_k = \text{MLP} \circ L_{k \rightarrow 0} \circ L_{k \rightarrow k}^{(T)} \circ \sigma \circ \dots \circ \sigma \circ L_{k \rightarrow k}^{(1)}, \quad (1)$$

함수 합성

where each $L_{k \rightarrow k}^{(t)}$ is equivariant linear layer [47] from $\mathbb{R}^{n^k \times d_{t-1}}$ to $\mathbb{R}^{n^k \times d_t}$, σ is activation function, and $L_{k \rightarrow 0}$ is a invariant linear layer from $\mathbb{R}^{n^k \times d_T}$ to $\mathbb{R}$.

- K-IGN은 이론적으로 k-WL test만큼 강력함.
- 2-IGN은 message passing GNN보다 표현력이 뛰어남.

Invariant Graph Network

- IGN의 핵심 구성 요소 : 노드 순열 대칭성을 존중하면서 최대 표현력을 갖는 equivariant linear layer

Definition 2. An equivariant linear layer is a function $L_{k \rightarrow l} : \mathbb{R}^{n^k \times d} \rightarrow \mathbb{R}^{n^l \times d'}$ written as follows for order- k input $\mathbf{X} \in \mathbb{R}^{n^k \times d}$:

$$L_{k \rightarrow l}(\mathbf{X})_{\mathbf{i}} = \sum_{\mu} \sum_{\mathbf{j}} \mathbf{B}_{\mathbf{i}, \mathbf{j}}^{\mu} \mathbf{X}_{\mathbf{j}} w_{\mu} + \sum_{\lambda} \mathbf{C}_{\mathbf{i}}^{\lambda} b_{\lambda}, \quad (2)$$

where $\mathbf{i} \in [n]^l, \mathbf{j} \in [n]^k$ are multi-indices, $w_{\mu} \in \mathbb{R}^{d \times d'}$, $b_{\lambda} \in \mathbb{R}^{d'}$ are weight and bias parameters, and $\mathbf{B}^{\mu} \in \mathbb{R}^{n^{l+k}}$ and $\mathbf{C}^{\lambda} \in \mathbb{R}^{n^l}$ are binary basis tensors corresponding to order- $(l+k)$ and order- l equivalence classes μ and λ , respectively. Invariant linear layer is a special case of $L_{k \rightarrow l}$ with $l = 0$.

레이어가 그래프의 동형성을 유지하도록 함

Can Self-Attention Approximate Equivariant Basis?

- Multihead self layer

$$\text{MSA}(\mathbf{X})_i = \sum_{h=1}^H \sum_j \alpha_{ij}^h \mathbf{X}_j w_h^V w_h^O \text{ where } \alpha^h = \text{softmax} \left(\frac{\mathbf{X} w_h^Q (\mathbf{X} w_h^K)^\top}{\sqrt{d_H}} \right), \quad (3)$$

where H is number of heads, d_H is head size, and $w_h^Q, w_h^K \in \mathbb{R}^{d \times d_H}$, $w_h^V \in \mathbb{R}^{d \times d_v}$, $w_h^O \in \mathbb{R}^{d_v \times d}$.

- 근사 가능

$$\alpha^2 = \text{softmax}(-a\mathbf{I}) \rightarrow \frac{1}{n-1}(\mathbf{1}\mathbf{1}^\top - \mathbf{I}) \text{ as } a \rightarrow \infty. \quad (6)$$

Pure Transformers are Powerful Graph Learners

- TokenGT는 최소한 k -IGN만큼의 표현을 가짐
- k -IGN은 k -WL test, GNN보다 강력함
- TokenGT는 최소한 k -WL test만큼 강력하고, GNN보다 표현력이 높음.

This directly leads to the following corollary:

Corollary 1. *A Transformer on node and type identifiers in Theorem 2 is at least as expressive as k -IGN composed of order- k equivariant linear layers.*

Corollary 1 allows us to draw previous theoretical results on the expressiveness of k -IGN [\[46, 47, 34\]](#) and use them to lower-bound the provable expressiveness of a standard Transformer:

Corollary 2. *A Transformer on node and type identifiers in Theorem 2 is at least as powerful as k -WL graph isomorphism test and is more expressive than all message-passing GNNs within the framework of Gilmer et al. (2017) [\[21\]](#).*

Approximating Second-Order Equivariant Basis

Table 1: Second-order equivariant basis approximation. We report average and standard deviation of L2 error averaged over heads over 3 runs. For Random/ORF (first-order), we sample random embeddings independently for each token.

node id.	type id.	dense input		sparse input	
		train L2 ↓	test L2 ↓	train L2 ↓	test L2 ↓
×	×	47.95 ± 0.600	53.93 ± 1.426	29.88 ± 0.450	34.70 ± 1.167
×	○	32.38 ± 0.448	40.06 ± 1.202	15.92 ± 0.275	20.39 ± 0.765
Random (first-order)	○	32.19 ± 0.476	32.49 ± 3.687	15.87 ± 0.247	16.56 ± 0.904
ORF (first-order)	○	32.35 ± 0.369	39.87 ± 1.263	15.87 ± 0.247	16.56 ± 0.908
Random	×	5.909 ± 0.019	5.548 ± 0.090	8.152 ± 0.042	8.270 ± 0.285
ORF	×	5.472 ± 0.035	5.143 ± 0.078	7.167 ± 0.025	7.190 ± 0.217
Laplacian eigenvector	×	1.899 ± 3.050	1.702 ± 2.912	0.288 ± 0.019	0.064 ± 0.010
Random	○	0.375 ± 0.009	0.234 ± 0.011	0.990 ± 0.108	0.875 ± 0.042
ORF	○	0.080 ± 0.001	$0.009 \pm 5e-5$	0.129 ± 0.002	0.011 ± 0.002
Laplacian eigenvector	○	$0.053 \pm 1.5e-5$	$0.005 \pm 1e-4$	0.101 ± 0.003	0.019 ± 0.007

Approximating Second-Order Equivariant Basis

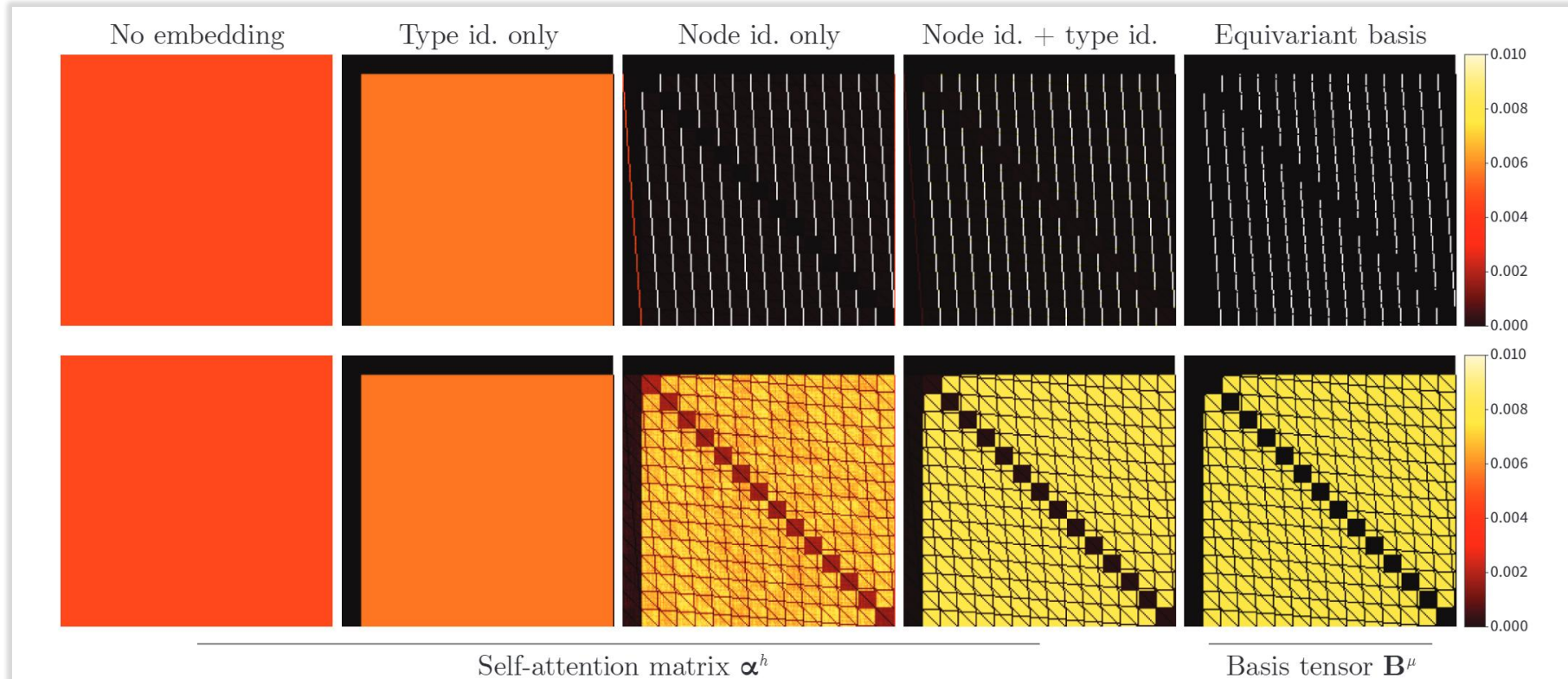


Figure 2: Self-attention maps learned under various node and type identifier configurations for two target equivariant basis tensors (out of 15). For better visualization, we clamp the entries by 0.01. Self-attention learns acute patterns coherent to equivariant basis when orthonormal node identifiers and type identifiers are both provided as input. More images can be found in Appendix A.4.1.

Large-Scale Graph Learning

Table 2: Results on PCQM4Mv2 large-scale graph regression benchmark. We report the Mean Absolute Error (MAE) on the validation set, and report MAE on the unavailable test set if possible.

method	# parameters	valid MAE ↓	test-dev MAE ↓	asymptotics
<i>Message-passing GNNs</i>				
GCN [27]	2.0M	0.1379	0.1398	$\mathcal{O}(n + m)$
GIN [27]	3.8M	0.1195	0.1218	$\mathcal{O}(n + m)$
GAT	6.7M	0.1302	N/A	$\mathcal{O}(n + m)$
GCN-VN [27]	4.9M	0.1153	0.1152	$\mathcal{O}(n + m)$
GIN-VN [27]	6.7M	0.1083	0.1084	$\mathcal{O}(n + m)$
GAT-VN	6.7M	0.1192	N/A	$\mathcal{O}(n + m)$
GAT-VN (large)	55.2M	0.1361	N/A	$\mathcal{O}(n + m)$
<i>Transformers with strong graph-specific modifications</i>				
Graphormer [63]	48.3M	0.0864	N/A	$\mathcal{O}(n^2)$
EGT [29]	89.3M	0.0869	0.0872	$\mathcal{O}(n^2)$
GRPE [54]	46.2M	0.0890	0.0898	$\mathcal{O}(n^2)$
<i>Pure Transformers</i>				
Transformer	48.5M	0.2340	N/A	$\mathcal{O}((n + m)^2)$
TokenGT (ORF)	48.6M	0.0962	N/A	$\mathcal{O}((n + m)^2)$
TokenGT (Lap)	48.5M	0.0910	0.0919	$\mathcal{O}((n + m)^2)$
TokenGT (Lap) + Performer	48.5M	0.0935	N/A	$\mathcal{O}(n + m)$

Large-Scale Graph Learning

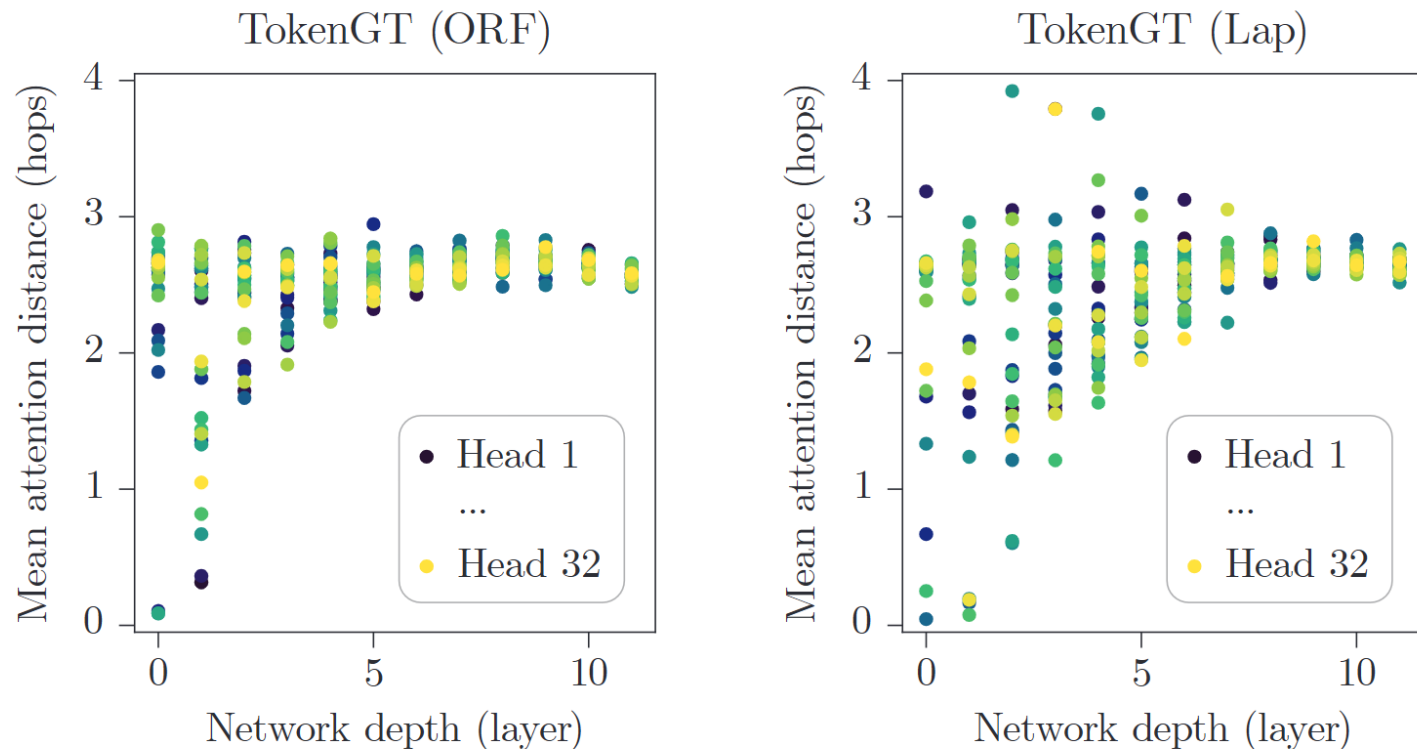


Figure 3: Attention distance by head and network depth. Each dot shows mean attention distance in hops across graphs of a head at a layer. The visualization is inspired by Dosovitskiy et al. (2020) [18]. More images can be found in Appendix A.4.2.

